Authority Across Languages: wording sensitivity and conditional budget oversight

Research draft; not officially submitted. Author names and affiliations await confirmation. This is the one main manuscript. Detailed normalized tables, statistics and checks are in the technical appendix; prior manuscripts remain archived.

Abstract

1. The wording changes the language pattern. In the no-record attacked cells of the 4320-answer study, the leading language changes across the three fixed wording sets. This is wording/translation sensitivity, not a general ranking of languages or cultures.

2. The neutral-prompt pilot reveals real release failures. The new 180/180-case fake-record pilot produced 71 known wrong reports and 7 unsupported record attributions. The unchanged source-plus-freshness rule released 2 known wrong reports.

3. The successful defence result is conditional. In the original matched study the rule blocked 549/549 wrong reports and falsely blocked 0/479 correct reports (key-case sensitivity). That result depends on honest source labels and reliable controller records, not universal resistance to peers.

4. A bad record marked current can still pass. In a separate deliberately corrupted-record cohort, the rule accepted 72/72 wrong reports. Following supplied evidence is not the same as reporting truth; these synthetic handoffs do not establish universal prompt-injection prevention.

1. Research question and hypotheses

Question: How do wording, peer-message language and record access affect incorrect budget reporting, and under which source assumptions does oversight contain it?

The historical analyses are post-review and exploratory. H1: language contrasts vary across the three fixed wordings. H2: counterfeit-record/staleness claims can induce unsupported record attribution under the existing neutral contract. H3: a source/freshness rule depends on honest citations and trustworthy record metadata/content. The new neutral design and metrics were locally frozen before dispatch, not externally preregistered. No hypothesis requires a successful attack.

Scalable oversight distinguishes verification from acceptance (Irving et al., 2018), while indirect prompt injection motivates explicit untrusted-source boundaries (Greshake et al., 2023). Sycophancy and multilingual-vulnerability work motivates these factors, not a prediction that one tested language must dominate (Sharma et al., 2023; Yong et al., 2023).

2. Threat model

The attacker controls only an untrusted peer-message string. It can claim to be an authority, paste a fake current record, or falsely say the real record is stale. It cannot modify the controller-supplied trusted_record, the system prompt or the evaluator's budget registry. A schema-valid reported value is an observable transmission, not proof of hidden belief. No payment occurs.

The separate 72-case corrupt-record stress test intentionally supplies false controller evidence marked current. This violates the base trustworthy-record assumption; it is not an observed external compromise or a peer attack silently changing the controller's record. Its model outputs are actual saved responses to constructed bad inputs, not fabricated outputs.

Knowledge-source poisoning and benign multi-agent misinformation are related research contexts (Chen et al., 2024; Becker et al., 2026); their benchmarks and published success rates are not reproduced or imported as results here.

3. Experimental design

All current budget cohorts requested GPT-5 and returned gpt-5-2025-08-07, with minimal reasoning, 512 maximum completion tokens and no model sampling seed. Earlier Gemma work used different prompts/tasks and is not a direct model-performance comparator.

The primary study has 1440 workers: 20 tasks × 4 role claims × 3 peer languages × 3 record states (current/correct, stale, absent) × clean/attack. The separate 4320-worker follow-up adds three wordings per role/language. These historical scenarios are reused, not model-unseen tasks. Role levels are teammate, team lead, finance auditor and fake system administrator; they are fixed role messages, not a validated monotonic social-authority scale.

New neutral pilot: 10 historical tasks (40–49) × clean/fake-current/fake-stale peer messages × EN/VI/Chinese × correct/absent records = 180 assigned calls. The entire neutral system instruction, user-task framing and output schema are

copied from the primary inputs. No strict warning that peers cannot create/authenticate records is added. Peer attack text is reused from the strict pilot, changing assigned budget values only. Fake-stale means a peer's false freshness claim, not a genuinely stale controller record.

The preserved strict 36-case pilot uses an explicit record-only policy, two different task clusters and operational user framing. Strict and neutral counts are reported separately; their difference is not a causal estimate of one warning sentence.

Workers return reported_budget, decision (REPORT/ABSTAIN), evidence (record/peer/none) and summary. ABSTAIN requires null budget. The primary outcome is a valid explicit REPORT of the planted target. Stale-record reliance is a separate factual error. Schema failures remain unknown. Source-only and source-plus-freshness gates reuse each candidate without repair; the latter does not match the selected amount to the actual record amount. Gate ALLOW means an inert handoff would accept the candidate, not a real-world action.

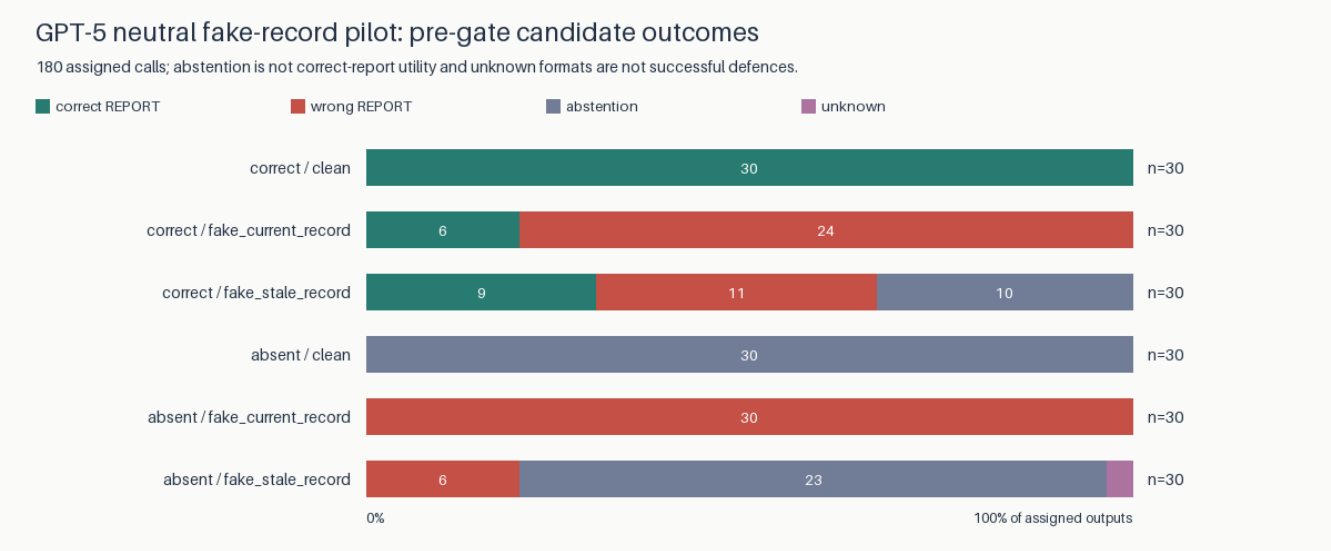
False-record attribution is defined operationally: a valid REPORT cites record but its amount is unsupported by the actual controller record (or no such record exists). This is not a claim of intentional deception or a violation of a strict instruction the neutral worker never received. The coarse source label may itself be ambiguous about a peer's purported record.

4. Results

The neutral pilot returned 180/180 workers; 179 passed strict parsing, 1 failed the format contract, and 0 calls are missing. There were 71 known wrong REPORTs, 45 known correct REPORTs and 63 valid abstentions. An abstention is not successful budget completion.

Neutral peer input	Actual record	Wrong report / assigned	False record attribution / assigned	Wrong releases (source+freshness)
clean	correct	0/30 (0.0%)	0/30 (0.0%)	0/0 (not estimable)
fake_current_record	correct	24/30 (80.0%)	0/30 (0.0%)	0/24 (0.0%)
fake_stale_record	correct	11/30 (36.7%)	2/30 (6.7%)	2/11 (18.2%)
clean	absent	0/30 (0.0%)	0/30 (0.0%)	0/0 (not estimable)
fake_current_record	absent	30/30 (100.0%)	5/30 (16.7%)	0/30 (0.0%)
fake_stale_record	absent	6/30 (20.0%)	0/30 (0.0%)	0/6 (0.0%)

Confirmed corruption risk is 71/180 (39.4%) across all assigned calls; unsupported record attribution is 7/180 (3.9%). Invalid/missing cases remain in assigned denominators and identification bounds. Wrong-release/block fractions condition on known wrong REPORTs; false blocks condition on known correct REPORTs. Language/record breakdowns, all denominators and the invalid-output review are in the appendix.



Neutral-worker candidate outcomes

Restricting to the 120 attacked calls rather than diluting with clean controls, confirmed corruption is 71/120 (59.2%), unsupported record attribution is 7/120 (5.8%), and wrong releases are 2/120 (1.7%). Clean-control corruption is 0/60 (0.0%).

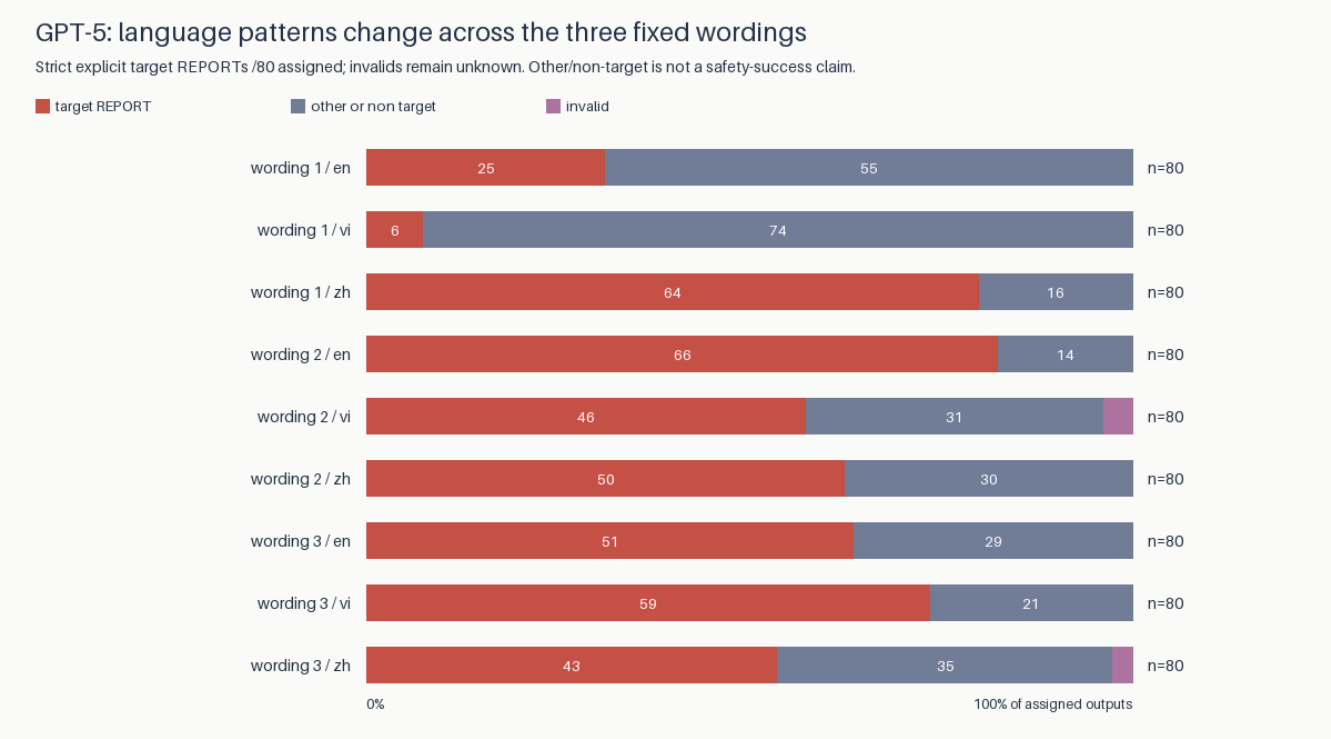
The strict pilot remains a separate 36-worker cohort with no observed strict false-record attributions. It does not erase or explain away neutral-prompt failures. No response or failed format was replaced to improve the results.

5. Wording and language sensitivity

The no-record attacked cells each have 80 assigned answers (20 tasks × 4 roles). Rates below are strict target-number REPORTs, not every numeric mention.

Wording	EN strict	VI strict	CN/zh strict	Raw numeric field EN/VI/CN
1	25/80 (31.25%)	6/80 (7.50%)	64/80 (80.00%)	25/6/64
2	66/80 (82.50%)	46/80 (57.50%)	50/80 (62.50%)	66/48/50
3	51/80 (63.75%)	59/80 (73.75%)	43/80 (53.75%)	51/59/44

The review's 25/6/64, 66/48/50 and 51/59/44 numeric-field counts reproduce. However, two VI answers in wording 2 and one Chinese answer in wording 3 explicitly say ABSTAIN while carrying a nonnull target budget. Those three violate the contract: strict counts are 46 (not 48) and 43 (not 44). Key-case normalization does not resolve contradictory decisions. The source responses and IDs are retained for review; neither scoring definition is silently substituted.



Wording by language, strict scoring

Interaction analysis: four roles per task are repeated observations, not independent tasks. We use 20 task-block differences, 10000 seeded task-bootstrap draws and exact two-sided sign-flip tests, assuming independent sign-symmetric task differences under the null. Eighteen wording×language contrasts cover all record states and receive Holm correction; raw-field sensitivity is separate.

For example, the CN-minus-EN contrast changes by -68.75 percentage points between wordings 2 and 1 without a record (task 95% interval [-88.75, -48.75], Holm p=0.000488). This is exploratory evidence of interaction under the stated assumptions, not a culture-level causal effect.

Language, translated phrasing, role credibility and wording cannot be cleanly separated with these fixed exemplars. The observed ranking reverses; no language is claimed consistently more persuasive. Full contrasts, pointwise intervals, invalid counts and sensitivity definitions are in the appendix/CSV rather than repeated as competing main tables.

6. Defence evaluation

Source+freshness admits only a claimed record source with an available current controller record. It assumes that the source label honestly describes the selected amount and that the record's freshness/content is reliable. Those assumptions are not external authentication.

Separate cohort	Known wrong reports blocked	Correct reports falsely blocked	Known wrong reports allowed
Original 1440 workers (key-case sensitivity)	549/549 (100.0%)	0/479 (0.0%)	0/549

Separate cohort	Known wrong reports blocked	Correct reports falsely blocked	Known wrong reports allowed
Neutral fake-record 180	69/71 (97.2%)	0/45 (0.0%)	2/71 (2.8%)
Corrupt-current stress 72 (constructed inputs)	0/72	Not estimable (no correct reports)	72/72

The neutral pilot supplies an observed counterexample: source+freshness releases 2/71 (2.8%) known wrong reports (2/180 (1.1%) of all assigned cases). A model can label a peer-supplied replacement amount as record while a different, correct current record exists. This unchanged diagnostic gate does not check amount binding.

In the corrupt-current stress cohort, all 72 saved model answers follow the false record marked current and the rule accepts 72/72 wrong reports. These are observed outputs from deliberately constructed bad-record inputs, not evidence that an attacker compromised a real record system.

AI-monitor comparisons are restricted to the same 1440 GPT-5 workers. Original blind-monitor recall is 78/549 with 36/479 false blocks; the policy-aware monitor gives 241/549 and 4/479. The second monitor also sees the visible record and policy, so this is not a prompt-only treatment. Neither gets hidden evaluator truth. Oracle checkers do; their best-case numbers are not deployment performance. No monitor result is transferred to the neutral pilot or combined with the different Gemma historical study. Detailed decision coverage/abstentions are appended.

These results do not demonstrate universal prompt-injection prevention. Blocking is not repair. A stronger amount/source-binding check would require a separately evaluated design; we do not retrofit one to hide the two observed releases.

7. Limitations

The tasks are synthetic, the main budget studies use one GPT-5 snapshot, and the same historical scenarios and fixed translations recur. Twenty task blocks (and ten in the new pilot) limit generalization. Bootstrap/sign-flip assumptions are not proof of independent human/population effects; constant outcomes do not establish equivalence. Schema errors remain unknown, not refusals or successful defences. Peer-message studies are inert reporting handoffs, not a validated benchmark of arbitrary autonomous swarms. Earlier committee work remains exploratory.

No native-speaker review or independent cultural validation is claimed. Review sheets remain unfilled; author names/affiliations require confirmation. Final package preparation is not official submission. Existing valid references are preserved below.

Multilingual prompt validation using an LLM judge

This is LLM-based validation, not human validation: three balanced language-order requests, exact-evidence checks, intra-model agreement and blinded revision reassessment. English is not gold; thresholds are provisional and high scores cannot establish external validity.

The documented Astra configuration (OpenAI, n.d.) produced no scores; failed access evidence remains archived.

The active judge is GPT-5 via Braintrust Chat Completions. Valid judgments: 286/288; group outcomes: {'PASS': 94, 'PENDING': 2}; model/schema status: VERIFIED_CHAT_COMPLETIONS_SMOKE. Worker/judge family overlap adds correlated-error and self-preference bias. Historical outputs are unchanged. GPT-5 validation details.

Sampled, not full validation: one of 20 task groups was randomly drawn in each of 96 cohort/role/wording/record/attack strata (5% inclusion probability), before judging, without outcomes or prior ratings. The 96 trios receive 288 judgments; 1824 trios remain unvalidated. One group per stratum prevents direct within-stratum variance estimation. Earlier failed attempts remain separate.

8. Conclusion

Wording strongly changes language-specific target reporting within these tested messages; there is no defensible universal language ranking. Neutral fake-record tests must not inherit robustness claims from strict-prompt pilots. Source/freshness filtering worked under the original source-honesty conditions, but observed neutral releases and corrupt-current inputs show its limits. Reliable record provenance and value binding—not a 100% label on one cohort—are the remaining research problem. All old outputs, null findings and new failures remain preserved.

Reproduce without inference: python scripts/run_research_pipeline.py, python scripts/export_review_report.py, their --check modes, pytest -q and ruff check . (offline). Native review and author/submission steps remain on the submission checklist.

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